I2C Bus

I2C Bus

C3 100nF +3.3V GND

U4 MCP23017
SDA 13 SCK 12 VDD 9 RESET 18 INTB 19 INTA 20 A2 17 A1 16 A0 15 VSS 10
GPB0 1 OSC1SAW
GPB1 2
GPB2 3 OSC1PULSE
GPB3 4
GPB4 5 OSC1SUB
GPB5 6
GPB6 7 OSC2_PW_SW
GPB7 8
GPA0 21 GLIDE
GPA1 22
GPA2 23 OSC1PW
GPA3 24
GPA4 25 OSC1PWM
GPA5 26
GPA6 27 OSC1_PW_SW
GPA7 28

C2 100nF +3.3V GND

U5 MCP23017
SDA 13 SCK 12 VDD 9 RESET 18 INTB 19 INTA 20 A2 17 A1 16 A0 15 VSS 10
GPB0 1 OSC2SAW
GPB1 2
GPB2 3 OSC2PULSE
GPB3 4
GPB4 5 OSC2TRIANGLE
GPB5 6
GPB6 7 OSC2_DETUNE_SW
GPB7 8
GPA0 21 FM_DEPTH
GPA1 22
GPA2 23 OSC2PW
GPA3 24
GPA4 25 OSC2PWM
GPA5 26
GPA6 27 FM_DEPTH_SW
GPA7 28

C1 100nF +3.3V GND

U6 MCP23017
SDA 13 SCK 12 VDD 9 RESET 18 INTB 19 INTA 20 A2 17 A1 16 A0 15 VSS 10
GPB0 1 KEYTRACK
GPB1 2
GPB2 3 FILTERLFO
GPB3 4
GPB4 5 FILTERATTACK
GPB5 6
GPB6 7 OSC1_PULSE_SW
GPB7 8
GPA0 21 FILTERCUTOFF
GPA1 22
GPA2 23 FILTERRES
GPA3 24
GPA4 25 FILTEREGLEV
GPA5 26
GPA6 27 OSC1_SAW_SW
GPA7 28

C4 100nF +3.3V GND

U7 MCP23017
SDA 13 SCK 12 VDD 9 RESET 18 INTB 19 INTA 20 A2 17 A1 16 A0 15 VSS 10
GPB0 1 OSC2DETUNE
GPB1 2
GPB2 3 OSC2INTERVAL
GPB3 4
GPB4 5 AMPATTACK
GPB5 6
GPB6 7 OSC2_SAW_SW
GPB7 8
GPA0 21 FILTERDECAY
GPA1 22
GPA2 23 FILTERSUSTAIN
GPA3 24
GPA4 25 FILTERRELEASE
GPA5 26
GPA6 27 OSC1_SUB_SW
GPA7 28

C5 100nF +3.3V GND

U8 MCP23017
SDA 13 SCK 12 VDD 9 RESET 18 INTB 19 INTA 20 A2 17 A1 16 A0 15 VSS 10
GPB0 1 LFO_RATE
GPB1 2
GPB2 3 LFO_DELAY
GPB3 4
GPB4 5 MODWHEEL_DEPTH
GPB5 6
GPB6 7 OSC2_TRI_SW
GPB7 8
GPA0 21 AMPDECAY
GPA1 22
GPA2 23 AMPSUSTAIN
GPA3 24
GPA4 25 AMPRELEASE
GPA5 26
GPA6 27 OSC2_PULSE_SW
GPA7 28

+3.3V R19 10K SDA i2c SDA
SCL i2c SCL
+3.3V 10K R20

C6 100nF +3.3V GND

U9 MCP23017
SDA 13 SCK 12 VDD 9 RESET 18 INTB 19 INTA 20 A2 17 A1 16 A0 15 VSS 10
GPB0 1 AT_DEPTH
GPB1 2
GPB2 3 EFFECT_MIX
GPB3 4
GPB4 5 VOLUME
GPB5 6
GPB6 7
GPB7 8
GPA0 21 PW_LFO
GPA1 22
GPA2 23 PITCHBEND_DEPTH
GPA3 24
GPA4 25 NOISE_LEVEL
GPA5 26
GPA6 27
GPA7 28

C7 100nF +3.3V GND

U10 MCP23017
SDA 13 SCK 12 VDD 9 RESET 18 INTB 19 INTA 20 A2 17 A1 16 A0 15 VSS 10
GPB0 1 PM_DCO2
GPB1 2
GPB2 3 PM_FILTERENV
GPB3 4
GPB4 5 AMP_LFO
GPB5 6
GPB6 7 NOISE_SW
GPB7 8
GPA0 21 EFFECTPOT1
GPA1 22
GPA2 23 EFFECTPOT2
GPA3 24
GPA4 25 EFFECTPOT3
GPA5 26
GPA6 27 EFFECT_MIX_SW
GPA7 28

Teensy4.1 U?
GND 1
MIDI IN 2 0_RX1_CRX2_CS1
MIDI OUT 3 1_TX1_CTX2_MISO1
DEMUX_EN 4 2_OUT2
ST7789_DC 5 3_LRCLK2
ENCODER A 6 4_BCLK2
ENCODER B 7 5_IN2
SR_595_DATA 8 6_OUT1D
SR_595_CLK 9 7_RX2_OUT1A
SR_595_LATCH 10 8_TX2_IN1
ST7789_RES 11 9_OUT1C
DAC_CS 12 10_CS_MQSR
DAC_SDA 13 11_MOSI_CTX1
14 12_MISO_MQSL
15 3V3
DCO_MIDI OUT 16 24_A10_TX6_SCL2
17 25_A11_RX6_SDA2
ST7789_SDA 18 26_A12_MOSI1
ST7789_SCL 19 27_A13_SCK1
20 28_RX7
DISPLAY_MIDI OUT 21 29_TX7
22 30_CRX3
23 31_CTX3
24 32_OUT1B
VUSB 49
(3.6v to 5.5v) VIN 48
GND 47
(250mA max) 3V3 46
23_A9_CRX1_MCLK1 45 SAVE
22_A8_CTX1 44 SETTINGS
21_A7_RX5_BCLK1 43 BACK
20_A6_TX5_LRCLK1 42 ENC_BUT
19_A5_SCL 41 I2C SCL
18_A4_SDA 40 I2C SDA
17_A3_TX4_SDA1 39
16_A2_RX4_SCL1 38
15_A1_RX3_SPDIF_IN 37
14_A0_TX3_SPDIF_OUT 36
13_SCK_LED 35 DAC_SCL
GND 34
41_A17 33
40_A16 32
39_MISO1_OUT1A 31
38_CS1_IN1 30 ST7789_CS
37_CS 29
36_CS 28 DEMUX 0
35_TX8 27 DEMUX 1
34_RX8 26 DEMUX 2
33_MCLK2 25 DEMUX 3
D− 66
D+ 67
USB Device
5V 55
D− 56
D+ 57
GND 58
GND 59
USB Host
R− 65
GND 64
T+ 63
T− 62
LED 61
R+ 60
Ethernet
VBAT 50
3V3 51
GND 52
PROGRAM 53
ON_OFF 54

+5V GND +3.3V GND

T4.1 PIN 49 165_DATA
T4.1 PIN 50 165_CLK
T4.1 PIN 51 165_LATCH

74HC165 shiftregisters on front panel
DATA PINS on Underside of Teensy 4.1

T4.1 PIN 54 AUTOTUNE_BUSY

Autotune Button
SW43
AUTOTUNE
2 1
68 R6 5 6
GND

SW?
RotaryEncoder_Switch
ENC_BUT S2 S1 B ENCODER B C A ENCODER A GND

SAVE 1 2 SAVE SW46
SETTNGS 1 2 SETTINGS SW47
BACK 1 2 BACK SW48
GND

+5V U? LM317_TO-220 VI 3 VO 2 ADJ 1 595_POWER
100uf C? 470 R? 330 R? C? 100n C? 10uf
75R R?
GND

MIDI IN 1 2 220 R? D? 1N4148 J?
U? 4N35 1 2 6 5 4
R? 100R R? 3k3
+3.3V GND MIDI_IN

J? MIDI OUT 1 2 3 +3.3V
33R R? 33R R? GND MIDI_OUT

J1 MIDI OUT 1 2 3 +3.3V
33R R1 33R R2 GND DCO_MIDI_OUT

J2 MIDI OUT 1 2 3 +3.3V
33R R3 33R R4 GND DISPLAY_MIDI_OUT

+3.3V J?
1 2 3 4 5 6 7 8
ST7789_SCL ST7789_SDA ST7789_RES ST7789_DC ST7789_CS
ST7789 Display

GND +3V3 C? 100n
U? DAC8568
LDAC 1 CLR 9 VREFIN 8 ADDR0 2 SCL 16 SDA 15 GND 14 AVDD 3
VOUTA 4 0_0_DAC
VOUTB 13 0_1_DAC
VOUTC 5 1_0_DAC
VOUTD 12 1_1_DAC
VOUTE 6
VOUTF 11
VOUTG 7
VOUTH 10
DAC_CS DAC_SCL DAC_SDA
+3V3 GND

Craig Barnes
Sheet: /
File: programmer.kicad_sch
Title: A Bit More Programmer
Size: A3 | Date: 2022-09-22 | **Rev: 1**
KiCad E.D.A. 9.0.0 | Id: 1/1